INTRODUCTION TO GENERATIVE AI

PART 1: WHAT IS ARTIFICIAL INTELLIGENCE (AI)?

Let's start with a **simple real-life question**.

When you open:

- YouTube → it suggests videos
- Amazon → it suggests products
- Gmail → it detects spam
- Google Maps → predicts traffic

How does the computer decide all this?

Because it is using: **Artificial Intelligence**

Definition:

Artificial Intelligence is:

"The ability of machines to perform tasks that normally require human intelligence."

Such as:

✓ Thinking
✓ Learning
✓ Understanding language
✓ Making decisions
✓ Problem solving

Example (Normal Programming vs AI)

Traditional Programming (What you do now)

You give: Input + Logic → Output

Example:

```
marks = 85

if marks > 50:
    print("Pass")
```

```
else:
    print("Fail")
```

Computer is **not intelligent** here. It is just following rules YOU wrote.

🤖 AI Programming

You give: Input + Output → AI learns Logic

Example:

You give:

Study Hours	Pass/Fail
1 hr	Fail
2 hr	Fail
5 hr	Pass
8 hr	Pass

AI learns:

"Oh! If study hours increase, chance of pass increases."

So now: model.predict(6 hours)

AI says:

✓ PASS ----→ Without you writing any rule!

This is **Artificial Intelligence**

PART 2: MACHINE LEARNING (ML)

AI has many subfields.

One important subfield is: **Machine Learning**

Definition:

Machine Learning means:

Computer learns from data without being explicitly programmed.

Instead of:

You writing logic,

Computer creates logic itself by **observing data**.

Example:

Spam Email Detection

You give:

Email Text	Spam/Not Spam
Win lottery now	Spam
Meeting at 5pm	Not Spam
Free money	Spam

ML learns pattern:

Words like "free", "lottery", "money" → Spam

Now new email comes: "You won free trip"

ML predicts: ✓ Spam

PART 3: DEEP LEARNING (DL)

Machine Learning has another advanced part: **Deep Learning**

Definition:

The way to make a computer "smart" by letting it learn from its own mistakes, just like a child does.

A way of teaching computers to learn by example, much like a human does. It is a specific type of Artificial Intelligence (AI) that uses "artificial neural networks"—layers of software inspired by the way neurons in the human brain work

Why Deep Learning?

Normal ML cannot handle:

❌ Images
❌ Voice
❌ Complex text
❌ Videos

Deep Learning can handle:

✓ Face Recognition
✓ Speech Recognition
✓ Language Translation
✓ ChatGPT

It uses:

Neural Networks (Inspired by Human Brain)

Just like our brain has neurons,

Deep Learning uses:

Artificial Neurons

PART 4: NOW COMES GENERATIVE AI

Traditional AI:

✓ Classifies
✓ Predicts
✓ Detects

But cannot:

✗ Create
✗ Write
✗ Design
✗ Speak

Generative AI: Can **CREATE NEW CONTENT**

Definition:

Generative AI is: AI that learns patterns from data and generates new data similar to it.

Example:

If AI reads 1 lakh emails,

Now you ask: "Write an email for leave request"

AI writes:

✓ Brand new email
✓ Human-like
✓ Grammatically correct

It is not copying. It is **generating**

Generative AI Can Create:

Type	Output
Text	Email, summary
Image	AI art
Code	Python code
Audio	AI voice
Video	AI avatar

PART 5: DISCRIMINATIVE VS GENERATIVE AI

Discriminative AI

Answers:

"What is this?"

Example:

Is this email:

✓ Spam
✗ Not Spam

Is this:

✓ Cat
✗ Dog

It only **classifies**

Generative AI

Answers:

"Create something new"

Example:

✓ Write email
✓ Generate report
✓ Create story
✓ Summarize session

Difference

Discriminative AI	Generative AI
Detects spam	Writes email
Predicts disease	Generates report
Classifies image	Creates image
Gives label	Gives content

PART 6: WHERE YOU WILL USE GEN AI AS A DJANGO DEV?

In your **MINDTHERA App**:

You currently do:

✓ Appointment booking
✓ Session notes
✓ Reports manually

With GenAI:

You can:

Feature	AI Use
Session Notes	Auto summary
Counselling	Suggest therapy
Student mood	Emotion analysis
Appointment	AI assistant
Reports	Auto generation

Example:

Student session note stored in DB:

```
notes = "Student feels anxious about exams"
```

Send to AI:

```
summary = generate_summary(notes)
```

AI returns:

✓ Therapy summary
✓ Suggested coping methods
✓ Risk analysis

- Store in PostgreSQL
- Show in dashboard

Now your Django app becomes: AI-Powered Web App

PART 7: GEN AI ECOSYSTEM (WHAT YOU WILL USE)

Step 1: Backend

✓ Django / Flask

Step 2: AI API

✓ OpenAI
✓ Gemini
✓ Claude

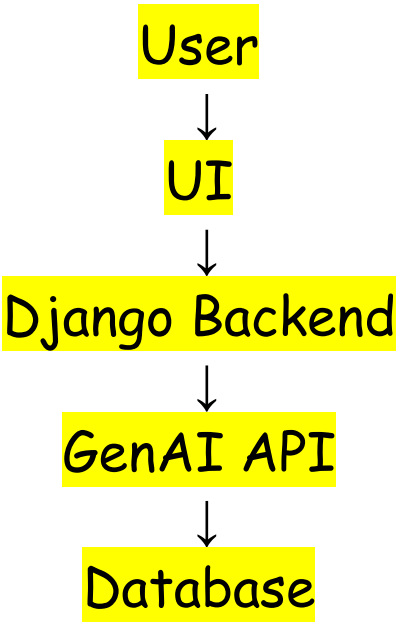
Step 3: Frontend

✓ React
✓ Angular

Step 4: Database

✓ PostgreSQL
✓ MongoDB
✓ SQlite3

Architecture:



PART 8: ROADMAP TO BECOME GEN AI ENGINEER

PHASE 1:

Python Basics
APIs
OOP

PHASE 2:

Machine Learning
Scikit-learn

PHASE 3:

Deep Learning
Neural Networks

PHASE 4:

Generative AI

Learn:

✓ Prompt Engineering
✓ LLM APIs
✓ Embeddings

PHASE 5:

Build Projects:

- AI Resume Analyzer
- AI Therapy Chatbot
- AI Financial Advisor
- AI Report Generator